=> Ex : 2.5

1) Reduce the following matrices into echelon form using elementary row operations:

-> echlon form:

$$\begin{bmatrix} 1 & A_{12} & A_{13} \\ 0 & 1 & A_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

(i) $$\begin{array}{c} \begin{bmatrix} 5 & 9 & 3 \\ -3 & 5 & 6 \\ -1 & -5 & -3 \end{bmatrix} \end{array} \begin{array}{l} R_1 \\ R_2 \\ R_3 \end{array}$$

1) Now we can interchange any Row.

2) Note we can multiply any Number with any R.
ex -> NR_n

3) $R_i + 2R_f$
(initial Row add or subtract with final Row.
-> Mean jis Row k liye hom ya jis row par hom operation perform kary gai us ko pehly likhy gai.

sol

By interchange R_3 to R_1

$$\begin{bmatrix} 5 & 9 & -3 \\ -3 & 5 & 6 \\ -1 & -5 & -3 \end{bmatrix} \quad R_3 \leftrightarrow R_1$$

$$\begin{bmatrix} -1 & -5 & -3 \\ -3 & 5 & 6 \\ 5 & 9 & 3 \end{bmatrix} \rightarrow (-1) R_1$$

$$\begin{bmatrix} 1 & 5 & -3 \\ -3 & 5 & 6 \\ 5 & 9 & -3 \end{bmatrix}$$

Now

$R_2 + -3R_1 \rightarrow R_2$

$$\begin{bmatrix} 1 & 5 & 3 \\ 0 & 20 & 15 \\ 5 & 9 & 3 \end{bmatrix} \quad \begin{matrix} -3 + 3(1) = 0 \rightarrow R_2 \\ 5 + 3(5) = 20 \rightarrow R_2 \\ 6 + 3(3) = 9 \rightarrow R_2 \end{matrix}$$

Now for $R_3 - 5R_1 \rightarrow R_3$.

$$\begin{bmatrix} 1 & 5 & 3 \\ 0 & 20 & 15 \\ 0 & 16 & -12 \end{bmatrix}$$

Now $\frac{1}{20} R_2$

$$\begin{bmatrix} 1 & 5 & -3 \\ 0 & 1 & 3/4 \\ 0 & -16 & -12 \end{bmatrix} \quad \text{Now } R_3 + 16R_2$$

$$\begin{bmatrix} 1 & 5 & -3 \\ 0 & 1 & 3/14 \\ 0 & 0 & 0 \end{bmatrix}$$

if last row come with zero, then it is also consider as echlon form.

Q2 Reduced the following matrices into reduced echlon form using elementry row operations:

(i) $\begin{bmatrix} 3 & 5 & 4 \\ 4 & 1 & 5 \\ 7 & 6 & 3 \end{bmatrix}$ for Reduce echlon form:-

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

sol

$\begin{bmatrix} 1 & 5/3 & 4/3 \\ 4 & 1 & 5 \\ 7 & 6 & 3 \end{bmatrix}$ $R_1 \frac{1}{3}$

Now $R_2 - 4R_1$

$\begin{bmatrix} 1 & \frac{5}{3} & \frac{4}{3} \\ 0 & -\frac{17}{3} & -\frac{1}{3} \\ 7 & 6 & 3 \end{bmatrix}$

$4 - 4(1) \to 0$

$\cancel{\frac{5}{3}}$ $1 - 4\left(\frac{5}{3}\right)$

$1 - \frac{20}{3}$

$\frac{3-20}{3} = \frac{-17}{3}$

$\begin{bmatrix} 1 & \frac{5}{3} & \frac{4}{3} \\ 0 & -\frac{17}{3} & -\frac{1}{3} \\ 0 & -\frac{17}{3} & -4 \end{bmatrix}$ $R_3 - 7(R_1) \to R_3$

Now make $-\frac{17}{3}$ to 1

$\begin{bmatrix} 1 & \frac{5}{3} & \frac{4}{3} \\ 0 & 1 & \frac{1}{17} \\ 0 & -\frac{17}{3} & -4 \end{bmatrix}$ $R_2 \to -\frac{3}{17}$

Now 1 above element make 0 and below element also.

→

$R_2 \to R_3 + \frac{17}{3} R_2$ | or $R_3 + \frac{17}{3} R_2 \to R_3$

$$\begin{bmatrix} 1 & \frac{5}{3} & \frac{4}{3} \\ 0 & 1 & \frac{1}{17} \\ 0 & 0 & -\frac{11}{3} \end{bmatrix}$$

$0 + \frac{1}{2}(0) = 0$

$-\frac{17}{3} + \frac{17}{3}(1) \to R_3$

Now above the 1 make 0.

$\frac{-17+17}{3} = 0$ ✓

$R_1 - \frac{5}{3} R_2$

$\frac{5}{3} - \frac{5}{3}(1) \to R_1$

$= 0.$

$\frac{4}{3} - \frac{5}{3}\left(\frac{1}{17}\right)$

$\frac{-1}{3}\left(\frac{1}{17}\right)$

$\frac{-1}{51}$

$-4 + \frac{17}{3}\left(\frac{1}{17}\right) \to R_1$

$-\frac{4}{1} + \frac{1}{3} = \frac{-12}{3} = -4$

$-\frac{4}{1} + \frac{17}{3}\left(\frac{1}{17}\right)$

$\frac{-12+17}{3}\left(\frac{1}{17}\right)$

$\frac{5}{3}\left(\frac{1}{17}\right)$

$\frac{5}{51}$

$$\begin{bmatrix} 1 & 0 & \frac{21}{17} \\ 0 & 1 & \frac{1}{17} \\ 0 & 0 & -\frac{11}{3} \end{bmatrix}$$

Now R_3 to 1.

$R_3 \to -\frac{3}{11} R_3$

$$\begin{bmatrix} 1 & 0 & \frac{21}{17} \\ 0 & 1 & \frac{1}{17} \\ 0 & 0 & 1 \end{bmatrix}$$

$-4 + \frac{17}{3}\left(\frac{1}{17}\right)$

$\frac{-12+1}{3}(1)$

$\frac{-11}{3}$

Now, Make R_1 to 0.

$$R_1 \to R_1 - \frac{21}{17}(R_2)$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{1}{17} \\ 0 & 0 & 1 \end{bmatrix}$$

Now $R_2 \to R_2 - \frac{1}{17}(R_3)$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\frac{1}{7} - \frac{1}{17}(1) = 0 \, R_2$

Hence done withe Redece Row echlon form.

→ 2.5

→ find the rank of the following matrices using elementry row operations.

i) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 0 & 2 & 2 \end{bmatrix}$

→ Note: for finding the rank of the matrix first we have to find echlon form of that matrix.
on that echlon from you can find rank.

ex:-
you find

$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$ → 1 rank beacuse row is non-zero
→ 2
→ 3

so rank is 3.

$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$ 1
2

here rank is 2.

Q4, Find the inverse of the following matrices using elementry row op.

→ i,
$$\begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{bmatrix}$$

$A^{-1} = ?$

first we find the inverse by Finding Adjoiny. but Now using row oper.

$[A : I]$

Unit Matrix :- whose diagonal 1 and other elements 0.

Note:- Matrix A and I order should be same.

when we solve then

$[A : I] \sim [I : A^{-1}]$